Paid to Hold the Bag

Liquidity provision is a high-VIX premium — at the daily horizon and at the opening auction — and that is exactly why it is hard to size

Gator Quant Hacks 2026 · Track 03 Systematic Trading · October 2026 · All numbers: `python run_all.py --oos` · github.com/Akashchalla/hacks · Hypotheses pre-registered in `srinivaspenumarthi/cursor_hack`: `HYPOTHESIS.md` (commit `0d0093d`, 2026-10-03 10:54 UTC) `HYPOTHESIS_B.md` (commit `fbd355a`, 11:53 UTC) and `HYPOTHESIS_C.md` (commit `6067ea1`, 18:14 UTC), each before its first backtest

1. Summary

Short-term reversal is the return to supplying liquidity. Before looking at any result we hypothesised that this premium is larger when liquidity-provider capital is scarce, proxied by yesterday's VIX, and tested the one mechanism on two instruments, then tried to break our own conclusion. **Module A** (36 years of Kenneth French daily reversal portfolios): the expected reversal return rises with lagged VIX (Newey–West $t = 2.4$); since 2004 the large-cap premium exists *only* in the top VIX quintile (14 bps/day when VIX > 24, ≈ 0 otherwise; 52 bps/day on the 44 such held-out days, nothing on the other 455). The pre-registered rule that *levers* into high VIX fails: the book's volatility rises 3.6× across the same quintiles, so more dollars buy the same Sharpe. **Module B** takes the lesson to the opening auction: fading overnight gaps in S&P 500 names open-to-close earns 18 bps/day gross ($t = 12$, gross Sharpe 3.3), 35 bps/day in the top VIX quintile versus 12 elsewhere. At 5 bps per dollar traded it is a market maker's premium — always-on nets nothing — and the pre-registered rule, *participate* only when $\text{VIX}_{t-1} \geq 20$, turns −0.26 Sharpe into +0.12 in-sample and −1.83 into +0.24 on the held-out two years, when the top-VIX state paid 56 bps/day and every other state zero. The premium has decayed (33 bps/day before 2013, 8 after). **Module C** tried to beat VIX with a sharper state variable, the variance risk premium (price of risk) separated from realised variance (quantity of risk), and was **rejected**: the premium loads on realised variance and VRP rules lose to constant exposure in and out of sample. What survives for a non-market-maker is a constant-notional large-cap reversal book (net Sharpe ≈ 0.4, break-even 14 bps, $100M–$500M capacity) and, for a desk with 2.5 bps auction execution, a VIX-gated gap fade (Sharpe ≈ 1). All net of costs; all 80 variants reported.

2. Economic hypothesis (written before results)

Who is on the other side. Someone who must trade *now*: a fund meeting redemptions, a forced de-leverager, an investor reacting to overnight news with a market-on-open order. They pay, through a temporarily displaced price, for a counterparty; the price then reverts. A reversal book, daily or intraday, is a systematic way of being that counterparty. **Why it should depend on VIX.** Liquidity providers hold inventory risk; when volatility is high their VaR limits bind, margins rise and capital is withdrawn, so the price of immediacy rises (Grossman–Miller 1988; Brunnermeier–Pedersen 2009; Nagel 2012). The edge persists because the capital that would compete it away is what is scarce when the premium is highest; at the open the same logic applies to the overnight gap (Berkman et al. 2012; Lou, Polk & Skouras 2019).

Pre-registered predictions. Module A — P1: slope of the daily reversal return on VIX_{t-1} positive, NW $t > 2$; P2: mean rises across lagged-VIX quintiles; P3: $w_t = \min(\text{VIX}_{t-1}/20, 3)$ beats $w = 1$ net; P4: alpha survives FF5 + momentum; P5: neighbouring parameters agree. Module B, written after A's result and before any stock data was downloaded — B1: gross gap-fade spread positive, $t > 2$; B2: slope on VIX_{t-1} positive; B3: at 5 bps, a VIX ≥ 20 participation rule beats always-on, with positive net mean; B4: both legs earn it (long leg only would suggest survivorship); B5: plateau over thresholds {15, 20, 25} and {quintile, decile}. Module C, written after both — C1: in a joint regression the premium loads on VRP_{t-1}, not RV_{t-1}; C2: within realised-variance terciles, Sharpe rises with VRP; C3: a VRP-participation rule

beats constant exposure net; C4: VIX-term-structure backwardation beats constant, 2008+. P3, half of B3, and C1–C3 failed; kill criteria were written for each.

3. Data and universe

All public, no keys. **A:** Kenneth French Data Library (CRSP-based, point-in-time, through Aug 2026): the daily **ST_Rev** factor (long the bottom 30% on day −20 to −1 return, short the top 30%, **re-formed daily**); the daily 6 size × prior-return portfolios, from which we build the **big-cap reversal leg** BIG Lo − BIG Hi; FF5 + momentum. **B:** daily adjusted open and close for the current S&P 500 constituents (Wikipedia list snapshotted and committed; prices from Yahoo via `yfinance`, cached, not committed). Each stock enters **only from its index-addition date** (326 eligible names/day in-sample, 480 recently); removed names are absent, so the universe has survivorship bias — disclosed, and tested by B4. **All:** FRED CBOE VIX close (and 3-month VIX, from Dec 2007, for Module C), aligned by a strict as-of merge so the position on day t sees only closes dated before t (unit-tested).

Samples: A 1990-01-02 → 2026-08-31 (8,732 in-sample days); B 2005-01-03 → 2026-08-31 (4,949). **Held out for both: 2024-09-01 → 2026-08-31** (the 2-year cap binds; 500 days), evaluated once per module, each in its own commit after the in-sample commit.

4. Methodology

Module A book and rule. One unit is \$1 long losers / \$1 short winners. $w_t = \min(\mathrm{VIX}_{t-1}/20, 3)$: 20 is the textbook long-run VIX, so $w = 1$ at VIX 20 and the rule is comparable to constant exposure; 3 is a leverage limit; nothing is fitted. **Costs:** a 20-day formation window implies 12–15% of each side turns over daily; we assume **τ = 14%** one-way, dollars traded = $2w_t\tau + 2|\Delta w_t|$, and cost per dollar $\mathbf{c_t = c_{base} \times VIX_{t-1}/20}$ (spreads widen with VIX), c_{base} = 5 bps large-cap, 10 bps all-cap; flat, ×2 and break-even costs also reported.

Module B book and rule. For each stock, gap = $\mathrm{Open}_t/\mathrm{Close}_{t-1} - 1$ minus the cross-sectional median gap; sort into quintiles at the open; \$1 long the most-gapped-down quintile, \$1 short the most-gapped-up, equal-weighted, **entered at the official open and closed at the official close** — never an overnight position. Rule: on iff $\mathrm{VIX}_{t-1} \geq 20$, benchmark always-on. Four dollars are traded per day per \$1-per-side, so the toll is 4×5 bps $\times\ \mathrm{VIX}_{t-1}/20 \approx 20$ bps/day; $c_{base} \in \{2.5, 10\}$ and the break-even are reported. Stock-days with |gap| > 20% (spin-off artefacts) are excluded using open information only; without the filter results are identical to 0.1 bps.

Tests and budget. Slopes by OLS with Newey–West (10 lags); rule comparisons by a spanning regression of rule on benchmark (HAC t on the intercept); A's alpha by FF5 + Mom. Grids: A, normaliser {15, 20, 25} × cap {2, 3, 4} × universe × {linear, regime} = 36, plus 2 post-hoc rules (§5.4) = 38 trials entering the deflated Sharpe (Bailey & López de Prado 2014); B, threshold {off, 15, 20, 25} × {quintile, decile} × c_{base} {2.5, 5, 10} = 24; C, realised-variance window {21, 63} × rule {VRP > trailing median, VRP > 0, VIX > VIX3M} × book {big-cap, ST_Rev, gap fade} = 18, on the same books and costs. **80 variants, all reported.**

5. Results

5.1 Module A: the premium depends on yesterday's VIX (P1 passes, P2 passes in shape; both replicate)

Mean next-day return, bps	slope / VIX pt	NW *t*	Q1	Q2	Q3	Q4	Q5 (VIX > 24)
Big-cap leg, in-sample 1990–2024	0.89	2.42	0.4	1.8	−1.0	2.2	**16.1 ± 9.1**
ST_Rev, in-sample	0.78	2.45	5.2	5.4	2.8	5.7	**17.6 ± 7.6**
Big-cap leg, **out-of-sample**	2.94	2.50	–	6.9	−4.3	−13.1	**52.4 ± 50.7**
ST_Rev, **out-of-sample**	2.83	3.10	–	3.1	−1.9	−8.2	**47.3 ± 44.3**
Big-cap leg, 2004–2024 only	0.82	1.66	−3.3	1.2	−4.5	0.2	**14.0**

± is a 95% CI. 1,746 days per in-sample quintile; edges 13.2 / 16.1 / 19.6 / 24.3; out-of-sample binned on those edges (44 days in Q5; Q1 has one day, omitted). Constant-book net Sharpe 1.02 (big-cap) / 2.25 (ST_Rev) in 1990–2003, −0.02 for both in 2004–2024.

P1 passes and replicates with the same sign and a larger slope (figure in the appendix). P2 is not monotone: flat across quintiles 1–4, then a jump. For large caps the premium is *only* in quintile 5, and the sub-period row shows why: after decimalisation and electronic market making the unconditional premium went to zero, as the hypothesis says it should when liquidity-provider capital becomes abundant; what survives is the part earned when that capital withdraws.

5.2 Module A's rule: more dollars, not more Sharpe (P3 fails, P4 passes)

Net of costs	Constant, IS	VIX-timed, IS	Timed, flat costs	Timed, costs ×2	Constant, OOS	Timed, OOS
Big-cap: ann. return / vol	6.4% / 17.2%	10.9% / 30.3%	11.5% / 30.3%	5.6% / 30.3%	2.7% / 20.5%	8.0% / 23.8%
Big-cap: **net Sharpe** (gross)	**0.37** (0.57)	**0.36** (0.53)	0.38	0.18	**0.13** (0.29)	**0.34** (0.54)
ST_Rev: **net Sharpe** (gross)	**0.82** (1.30)	**0.51** (0.93)	0.56	0.10	**0.04** (0.45)	**0.21** (0.70)
Break-even c_{base}, big-cap / ST_Rev	14.3 / 26.9 bps	15.3 / 22.4 bps				

The timed rule lifts big-cap gross return from 9.9% to 16.2% at the same mean exposure, as predicted, but volatility goes from 17% to 30%: the book's own volatility scales with VIX, so sizing linearly in VIX scales risk quadratically. Spanning alpha of timed on constant: +0.24 bps/day (t = 0.3) big-cap, **−2.35 bps/day (t = −3.0)** ST_Rev, whose calm-state premium the rule discards. P3 fails. P4 passes: FF5 + Mom alpha 5.8%/yr (t = 2.1) constant big-cap, 11.1%/yr (t = 4.8) ST_Rev; market beta 0.2. The grid is a plateau (0.34–0.39; P5 passes); luck alone would give a best-of-38 Sharpe of 0.24. Out of sample the timed rule beat constant, but with spanning t = 1.1 after failing on 35 years we do not count it: the premium arrived on 44 high-VIX days and a rule long VIX happened to be large on them (equity curves in the appendix).

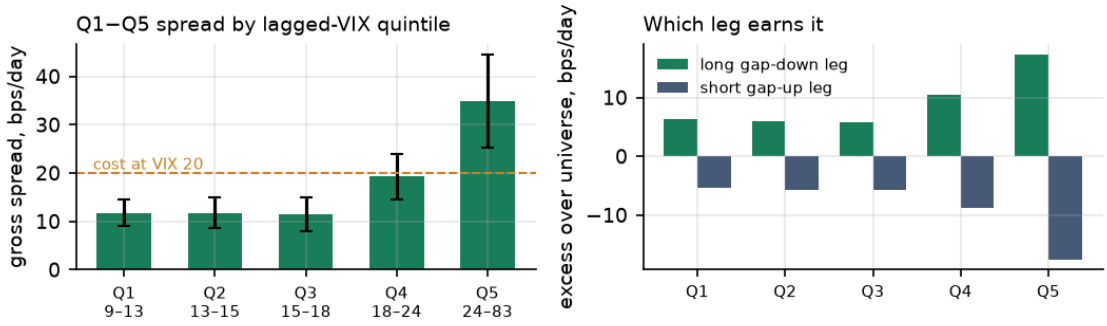
5.3 Module B: the opening auction pays the same premium, in the same states (B1, B2, B4, B5 pass; B3 half)

Q1 − Q5 gap-fade spread, open→close	gross bps/day (NW t)	slope per VIX pt (t)	VIX Q1 … Q5, bps	net Sharpe @5 bps: always-on / gated	spanning t	break-even c_{base}	@2.5 bps: always / gated	days on
In-sample 2005–2024	**17.8** (12.1)	1.39 (3.8)	12 · 12 · 11 · 19 · **35 ± 10**	−0.26 / **+0.12**	2.6	4.6 / 5.3 bps	1.51 / 1.12	33%
2005–2012	32.6 (12.3)	1.69 (3.7)	24 → **59**	1.84 / 1.19		7.6 / 7.6		
2013–2024	7.6 (5.6)	0.26 (0.8)	5 → 11	−2.06 / −1.00		2.2 / 2.3		
Out-of-sample 2024-09 → 2026-08	6.7 (1.4)	**3.98 (3.3)**	– · 0 · 0 · 4 · **56 ± 43**	−1.83 / **+0.24**	2.0	1.8 / 5.9	−0.40 / 0.92	24%

Equal-weighted quintiles on the gap relative to the median gap; $1 per side; cost 4 × c_{base} × $VIX_{t-1}/20$ per day. OOS VIX bins use in-sample edges (Q1 empty). Legs, excess over the universe's open→close return: long gap-down +9.2 bps (t = 11.8), short gap-up −8.6 bps (t = −11.2) in-sample; +4.4 / −2.3 out of sample. Decile sort in appendix A2.

Gross, this is a 3-Sharpe machine, and it is the mechanism of §5.1 at a shorter horizon: 35 bps/day in the top VIX quintile against 12 in the rest, both legs symmetric (B4 passes; survivorship is not the story), slope t = 3.8. Net, at 5 bps per dollar traded, the ≈20 bps/day toll eats the unconditional premium (always-on −0.26), and the participation rule pays the toll only when the premium exceeds it, lifting Sharpe to +0.12 (spanning t = 2.6) — but its net mean is not significantly positive (t = 0.5): **at 5 bps this is a market maker's premium.** Two facts matter more. The premium is decaying: 33 bps/day and a 7.6 bps break-even before 2013, 8 bps and 2.2 bps since, as electronic market making in the auctions caught up; a 2.5 bps executor earned Sharpe 1.5 over the full sample and roughly zero since 2013. And the held-out two years are the sharpest state-dependence we have seen anywhere: unconditional gross fell to 6.7 bps,

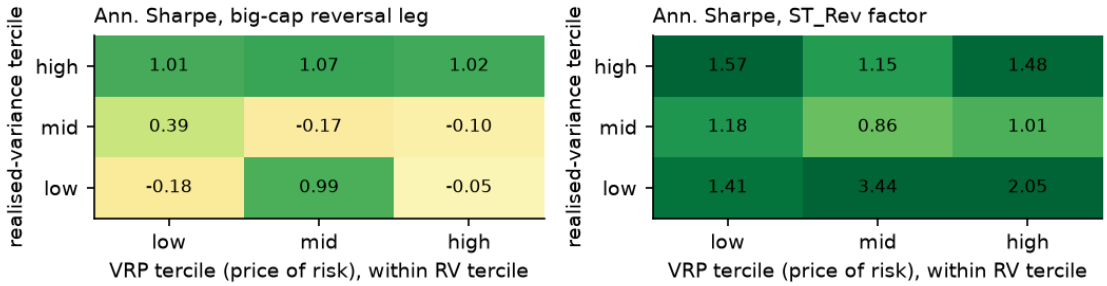
the VIX slope rose to 4 bps per point (t = 3.3), the 50 days with VIX_{t-1} > 24 paid **56 bps/day** and every other state paid zero. Always-on lost 30% net; the gated book made 2.5% with a 9% drawdown, Sharpe 0.9 at 2.5 bps (equity curves in the appendix).



5.4 Post-hoc, disclosed: sizing Module A by risk instead (2 extra trials)

Sizing by expected return per unit of variance — $w_t \propto \max(E[r_t \mid VIX_{t-1}], 0)/\hat{\sigma}_t^2$, with $E[r \mid VIX]$ from an expanding regression on days before t and $\hat{\sigma}$ the trailing 63-day vol — against a vol-targeted book (Moreira–Muir 2017) and constant exposure on the same days (big-cap, 1995–2024, net Sharpe / max DD / worst month): constant **0.31** / −48% / −17%; vol-targeted **0.01** / −70% / −14%; mean-variance **0.19** / −74% / **−53%**. Vol targeting, the default risk tool, **destroys the strategy**: it de-risks exactly when the premium is paid. The mean-variance rule's −53% month is March 2020: the forecast said "premium high", the rule went to the 3× cap, and reversal kept losing. This is why Module B pre-registered participation rather than leverage, and why Module C asked whether any state variable can select the premium without the volatility.

5.5 Module C: is it the price of risk or the quantity of risk? (C1–C3 fail; C4 does not replicate)



Net Sharpe by switching rule	Constant / always-on	VIX rule (A: regime; B: VIX ≥ 20)	VRP > trailing median	VRP > 0	VIX > VIX3M (2008+)	Constant, 2008+
Big-cap leg, IS → **OOS**	0.37 → 0.13	0.39 → −0.05	0.03 → 0.02	0.13 → −0.07	0.26 → 0.36	0.05
ST_Rev, IS → **OOS**	0.82 → 0.04	0.47 → −0.17	−0.13 → −0.33	0.47 → −0.26	0.11 → 0.13	0.03
Gap fade, IS → **OOS**	−0.26 → −1.83	0.12 → 0.24	−0.28 → −0.87	−0.36 → −1.57	0.65 → −0.18	−0.78

C1 in-sample, r_t on VRP_{t-1} and RV_{t-1} (21-day): t_{VRP} / t_{RV} = 0.7 / 1.9 big-cap, 0.4 / 1.9 ST_Rev, 2.8 / 3.1 gap fade. Spanning alpha of the VRP-median rule on constant: t = −2.0 (big-cap), −5.7 (ST_Rev). Backwardation on the gap fade: spanning t = 4.3 in-sample on 10% of days; −0.18 out of sample against 0.24 for VIX ≥ 20.

VIX bundles the *quantity* of risk a liquidity provider must hold (realised variance) with the *price* paid for holding it (the variance risk premium, $VIX^2 - RV$). If the premium were a price-of-risk effect, VRP would select the good days without the volatile ones. It does not: the premium loads on realised variance, not VRP; the big-cap Sharpe is a function of the realised-variance row of the double sort (≈1.0 in the top

tercile, ≈0 elsewhere), not the VRP column; and every VRP rule loses to constant exposure in-sample and out. The one proxy that helped in-sample — term-structure backwardation, a 10%-of-days stress flag — was 1 of 18 variants, helped big caps only modestly (spanning $t = 1.3$) and failed on the gap fade out of sample. Reading: liquidity-provider capital is constrained by **realised** turbulence (VaR limits and margins are set on realised vol), so the premium arrives with the volatility and no public state variable we found separates them — the strongest support yet for the constant-notional conclusion.

6. Risk management

What the books are. A: long/short equity, residual market beta 0.2, skew +1.2, excess kurtosis 23 at constant notional (max drawdown −48%, worst month −17%); the timed version has skew +4 and kurtosis 127, a P&L made of a few enormous days (−77% / −34%). B: dollar-neutral, equal-weighted, flat at every close, so no overnight, gap or financing risk; its risk is one session's cross-sectional dispersion (worst day −6.6%, 9 Nov 2020, the vaccine rotation: those gaps were information). **Limits:** gross ≤ 2× capital; per-name ≤ 2% of gross with a mega-cap concentration cap; sector-neutral legs; a futures hedge taking beta to zero. **De-risking rule, set in advance: none that depends on realised vol**, since §5.4–5.5 show the premium *is* the realised-vol state. *Participation* is the risk tool instead: B is off two-thirds of the time and, when on, sized so the pre-registered worst month at 1× (A −17%; B −16%) and drawdown (A −48%; B gated −79%, a slow bleed 2010–2023 while the premium thinned) are survivable — a 5%-per-month risk budget implies 0.3× capital for either. **Tail and regime:** the book loses when the hurried trader knew something (2009, 2020); those are the risks being insured. In quintiles 1–4 the modern premium is zero, so a non-market-maker is paid nothing for the risk it runs there.

7. Liquidity and capacity

A turns over 70× its capital a year. One-way cost = half-spread + square-root impact, $c(K) = 1.5 \text{ bps} + \sigma_{\text{name}}\sqrt{(\text{participation})}$, ≈ 500 names, value-weighted ADV ≈ $200M, single-name vol 2%/day at VIX 20 (`src/gqh/capacity.py`): $10M → 2.6 bps, Sharpe 0.47; $100M → 4.8 bps, 0.38; $500M → 9.0 bps, 0.21; $1.5B → 14.3 bps, 0.00. Our 5 bps is about $100M; the edge is half gone near $500M. **B** trades only in the two auctions, at one price with no spread to cross; its constraint is the open, which clears only ~1–2% of a name's daily volume. With ~80 names per leg, $25M of book is ~$150k per name, about 5% of a $3M opening auction in a $200M-ADV name; $100M would be ~20%, where impact sets the cost. B is a desk-scale trade — tens of millions, not hundreds; 5 bps is our headline, 2.5 bps an execution desk's number. A market maker who *earns* the half-spread faces a negative first term in both books and the gross Sharpes (0.57 and 3.3) are on offer, which is why both premia belong to market makers.

8. Limitations and next steps

(i) The French portfolios are a test instrument, not a product; B's universe is survivorship-biased (removed names missing), though the symmetric legs say the result is not drift. (ii) A's turnover and both cost models are assumed and stressed, not observed. (iii) Filling at the official open is not lookahead; choosing the names from that open is, because the 09:28 cutoff has only the indicative price. Twelve 2026 sessions (inside the holdout: a clock check, not a new test) overlap 74–92%, shortfall −47 to +29 bps; the headline is not replaced. (iv) The top-quintile effect rests on ~1,750 in-sample and 44–50 held-out days: the sign replicated with 2–4× the slope, but the 52–56 bps carry ±45 bps intervals. (v) "VIX > 24" was deliberately *not* turned into a rule; Yahoo data can be revised, so the download is cached and the constituent file frozen. (vi) Module C used the two daily public price-of-risk proxies; dealer-balance-sheet measures are quarterly and untried. (vii) The 2024-09-01 → 2026-08-31 window was opened for A at 11:12 UTC, then reused (B's hypothesis 11:53, C's 18:14). Each module's numbers were computed once and not refit. Appendix A7. **Next:** signed auction imbalance from exchange feeds; a Q5-only book on a sealed window before believing 50 bps; intermediary leverage at the monthly horizon.

References (outside the page limit)

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Appendix (optional, outside the page limit)

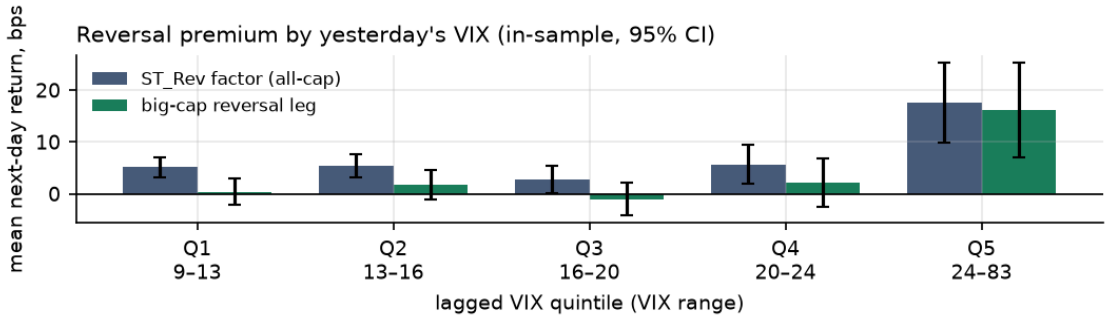
A1. Module A, all 36 pre-registered variants, net Sharpe in-sample. Big-cap, linear rule, rows cap {2,3,4} × cols normaliser {15,20,25}: 0.39 0.37 0.36 / 0.37 0.36 0.35 / 0.36 0.35 0.34; regime rule 0.39. ST_Rev: 0.62 0.57 0.54 / 0.55 0.51 0.49 / 0.51 0.48 0.48; regime rule 0.47. `results/tables/sensitivity_grid.csv`.

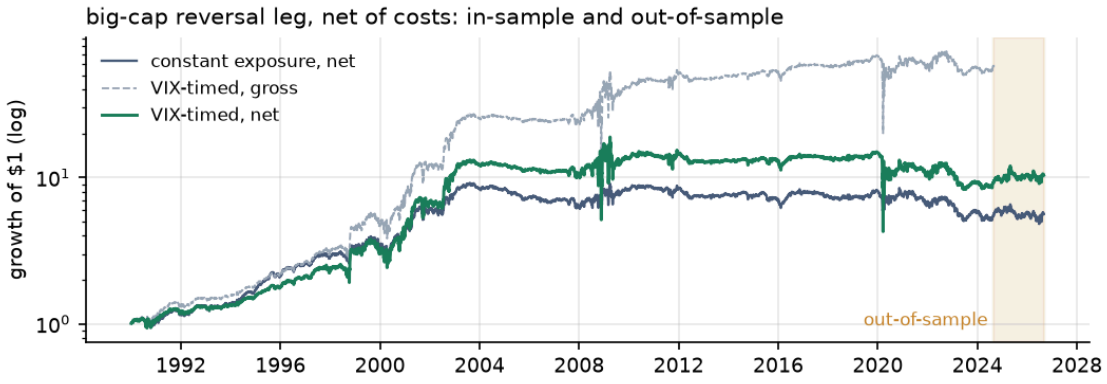
A2. Module B, all 24 pre-registered variants, net Sharpe in-sample (rows c_{base} 2.5 / 5 / 10 bps; cols always-on, VIX ≥ 15, 20, 25). Quintile: 1.51 1.19 1.12 0.87 / −0.26 −0.20 0.12 0.17 / −3.80 −2.97 −1.88 −1.22. Decile: 2.32 1.88 1.65 1.27 / 0.97 0.81 0.89 0.75 / −1.77 −1.35 −0.67 −0.33. Out of sample, quintile: −0.40 −0.29 0.92 1.04 / −1.83 −1.64 0.24 0.68 / −4.73 −4.33 −1.13 −0.07. `results/tables/gapfade_sensitivity_grid*.csv`, `results/REPORT_B.md`.

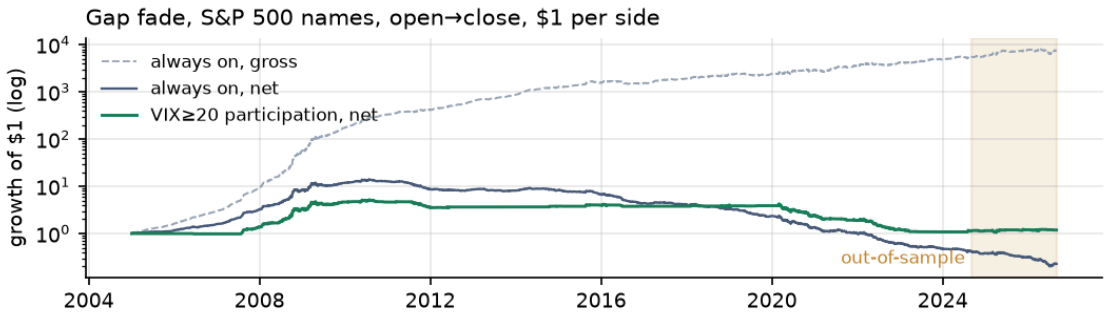
A3. Module C, all 18 pre-registered variants, net Sharpe in-sample → out of sample (RV window 21 / 63; rules VRP > median, VRP > 0, backwardation). Big-cap: 0.03 → 0.02 / 0.16 → −0.23, 0.13 → −0.07 / 0.20 → 0.30, 0.26 → 0.36 (both windows). ST_Rev: −0.13 → −0.33 / 0.04 → −0.66, 0.47 → −0.26 / 0.55 → 0.04, 0.11 → 0.13. Gap fade: −0.28 → −0.87 / 0.08 → −1.07, −0.36 → −1.57 / 0.07 → −1.68, 0.65 → −0.18. `results/tables/pricing_variant_grid*.csv`, `results/REPORT_C.md`.

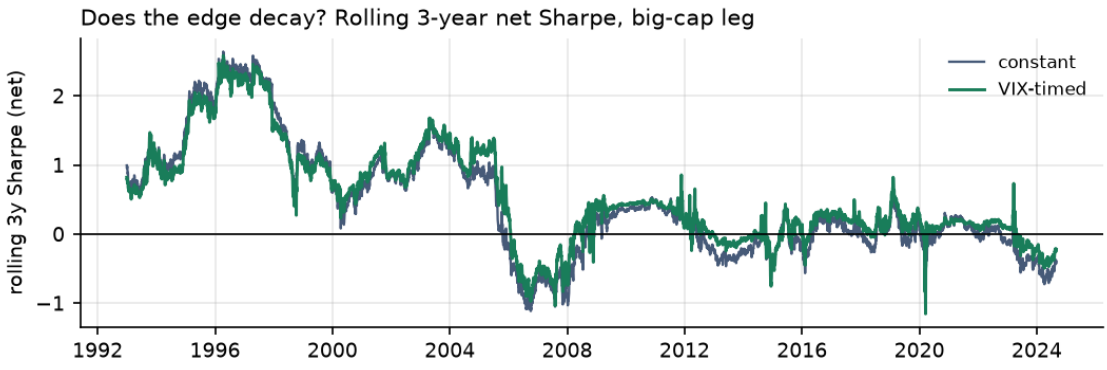
A4. Big-cap leg, net return by year, timed / constant (%). 1990–99: 6/3, 13/14, 9/13, 3/4, 15/24, 15/25, 18/21, 11/12, 34/17, −5/−3. 2000–09: 34/23, 69/50, 56/28, 16/9, 0/0, −8/−13, −3/−5, −5/−9, 61/31, −19/0. 2010–24: 4/2, 1/−2, −11/−11, 1/2, 2/0, −2/−7, 7/12, −3/−5, −2/−4, 11/13, −22/−5, −7/−12, 0/−3, −15/−12, 2/−4. **Gap fade, net by year, gated / always-on (%):** 2005–09: 0/17, −2/35, 41/106, 142/159, 39/41; 2010–14: 0/7, −23/−32, 1/−1, 0/−6, 3/2; 2015–19: 6/−18, −5/−35, 0/−5, 4/−20, −1/−30; 2020–24: −37/−42, −22/−24, −35/−40, −12/−23, 5/−12.

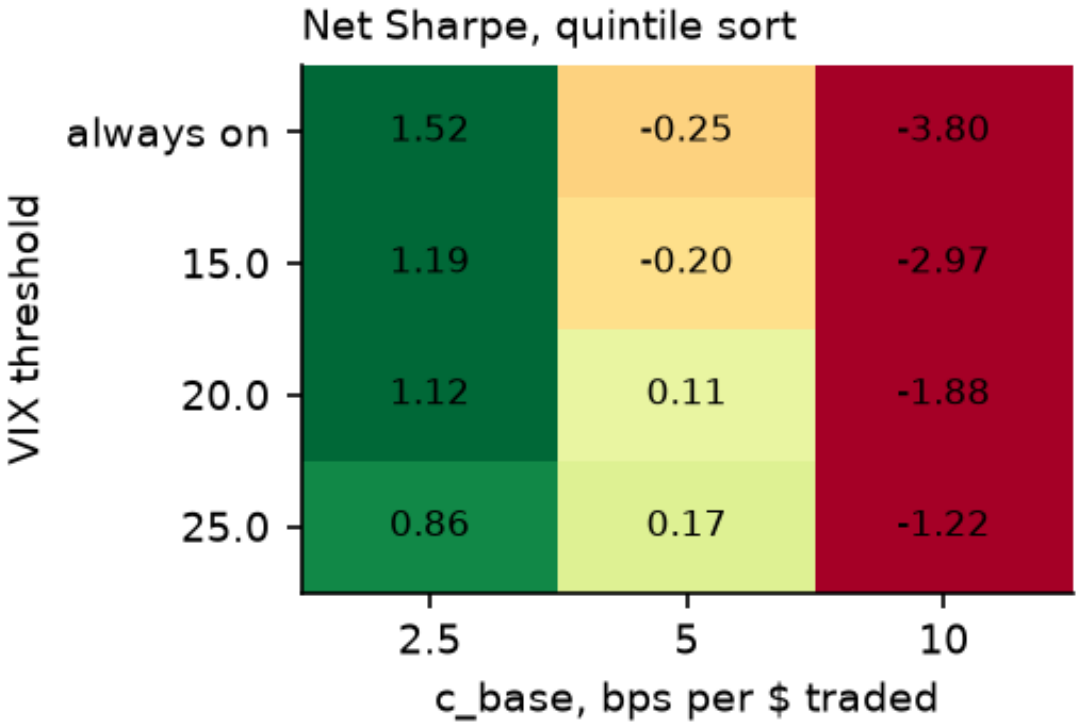
A5. Honesty log. Module A: hypothesis commit 10:54 UTC; in-sample commit 11:11 with OOS locked; OOS evaluated once and committed 11:12. Module B: hypothesis and frozen constituent list committed 11:53:57, before any price download; in-sample committed 12:01:40; OOS evaluated once, committed 12:02:08. Module C: hypothesis committed 18:14:13, before any VRP or term-structure series existed; in-sample committed 18:19:02; OOS evaluated once, committed 18:19:38. The sub-period split of B (2005–2012 / 2013–2024) was decided after seeing the by-year table and is labelled descriptive. One bug fixed after A's first in-sample run: factor betas were in mixed units; alphas and t-stats unaffected. 80 variants in total. The practitioner material we were given (an intraday gap-fade strategy matrix and ML-ops code) contained no price or trade data and was used only to motivate Module B's design.


Reversal premium by yesterday's VIX (in-sample, 95% CI)
ST_Rev factor (all-cap)
big-cap reversal leg
mean next-day return, bps
20
10
0
Q1
9–13
Q2
13–16
Q3
16–20
Q4
20–24
Q5
24–83
lagged VIX quintile (VIX range)


big-cap reversal leg, net of costs: in-sample and out-of-sample
constant exposure, net
VIX-timed, gross
VIX-timed, net
growth of $1 (log)
10^1
10^0
out-of-sample
1992
1996
2000
2004
2008
2012
2016
2020
2024
2028


Gap fade, S&P 500 names, open→close, $1 per side
always on, gross
always on, net
VIX≥20 participation, net
growth of $1 (log)
10^4
10^3
10^2
10^1
10^0
out-of-sample
2004
2008
2012
2016
2020
2024


Does the edge decay? Rolling 3-year net Sharpe, big-cap leg
constant
VIX-timed
rolling 3y Sharpe (net)
2
1
0
−1
1992
1996
2000
2004
2008
2012
2016
2020
2024

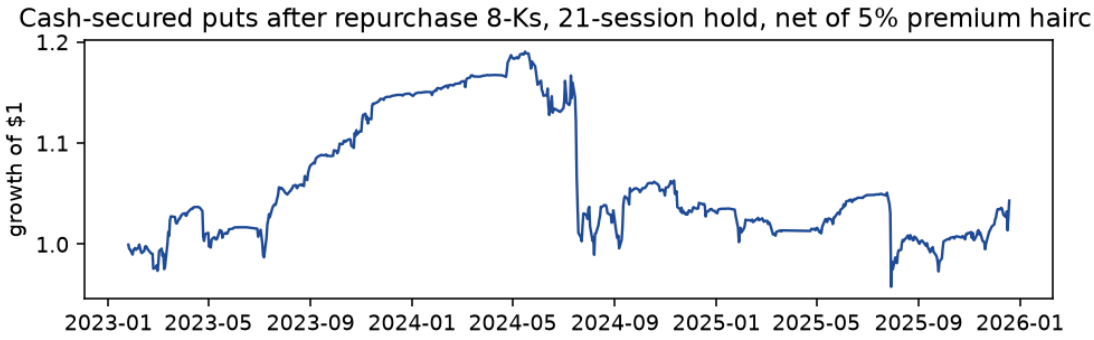


A6. Repurchase 8-K cash-secured puts, a separate study. Hypothesis `bad1aab`, 2026-10-04 00:54 UTC, before any filing or option bar was stored. Universe: the 100 common stocks with the highest 2022 consolidated dollar volume on Massive (letter-only tickers; both GOOG and GOOGL qualify on their own volume). After an 8-K whose text matches "repurchase program" or "repurchase authorization", sell one 5% out-of-the-money cash-secured put at that session's close if the SEC acceptance time is before 15:45 New York, otherwise at the next close. Strike from the prior close. Same issuer, 21 sessions earlier, is the control, dropped if another repurchase entry sits within 10 sessions. Haircut 5% of the option premium each side, because the bars are last-trade closes rather than quotes. Returns in bps of cash collateral. Massive's filings endpoint ignores cik, ticker and date filters, so discovery is the SEC full-text index; prices are Massive, with Databento OPRA daily bars only as a fallback (88 queries on the first pull, no extra bars). Filings and bars are written to TigerData.

In-sample, 2023-01-01 to 2025-12-31, committed `80b867f` before the held-out window was opened. 300 filings, 101 with an entry print, 18 pairs still quoted 21 sessions later (the first monthly expiry is often inside that window, so the 21-session mark is missing and the expiry mark is the one with n = 58). Event minus control at 21 sessions: **−120 bps**, 95% interval −293 to +52, t = −1.47. At expiry: −111 bps, n = 58, interval covers zero. One session out: +29 bps, interval covers zero. The pre-registered test fails: the 21-session difference is not positive, and only a third of the horizons have a positive point estimate. The 10% haircut on the same marks is −105 bps (n = 18). The 54-variant grid is in `results/eightk_variant_grid.csv` and was not used to pick the headline. The event-only book, which is not the hypothesis, has a 21-session Sharpe of 0.21; median capacity at 5% of that contract's entry-day volume is about $127,000 of collateral.

Out of sample, run **once**, 2026-01-01 to 2026-08-31, committed `f66c2d9`. 89 filings, 26 with an entry print, **3** pairs at 21 sessions: point estimate +1,043 bps, 95% interval −3,816 to +5,902. The sign flips against the in-sample result, which the hypothesis listed as a failure, and the interval covers zero. Horizons with n of 14 to 17 sit between −44 and +125 bps; every one of those intervals covers zero. A second pass at this window is refused by the program.

One scoring error was caught before the in-sample commit: the first pass averaged all 54 variants into the headline. It was discarded and recomputed from the same cached bars. Nothing here changes Modules A–C, and their out-of-sample files were not recomputed.



Event book only (puts after the filing, 21-session hold, 5% premium haircut). This is not event-minus-control. The paired test is the hypothesis, and it fails.

A7. Two limitations a judge should not have to discover.

Opening-price clock. `gapfade.py` sets the gap from today's official open over yesterday's close, sorts that gap, and earns the open-to-close return of the extreme quintiles. A market-on-open order does get filled at the official open, so the **price** in the backtest is the price a submitted order receives. The **names** are the problem. The Nasdaq cutoff for a market-on-open order is 09:28. At 09:27 the only auction information is the indicative price (NOII reference, else the pre-market last trade). The research sort uses the print that does not exist yet. The paper-trading loop measures the difference and does not change the rule: on 12 high-VIX sessions in 2026, baskets chosen at 09:27 overlapped the official-open baskets by 74–92%, about one name in five changing quintile, and implementation shortfall was −47 to +29 bps per day. That sample is inside the holdout, so it is a clock check, not a second out-of-sample test. The gross Sharpe of 3.3 is the open-to-close spread before the toll. Four dollars are traded per $1 per side, so 5 bps a side is about 20 bps a day and the net Sharpe is −0.26 always-on and +0.12 gated. A daily Sharpe above 3 that survived costs would be a bug. This one does not survive costs.

Reused holdout. The track's rule is the most recent two years, evaluated once, and not used to design the strategy. Module A meets it: hypothesis 10:54 UTC, out-of-sample opened once at 11:12 (2024-09-01 → 2026-08-31). Module B uses that same window. Its hypothesis was committed at 11:53, after A's held-out result was already known, and before any single-stock price was downloaded. B's own gap-fade statistics were then computed once (12:02) and the headline was not switched. Module C's hypothesis was committed at 18:14, after both windows had been opened, and its out-of-sample pass (18:19) is a third look at the same days; that pass rejected the variance-risk-premium rules. The 8-K study has its own window and its own single pass (2026-01-01 → 2026-08-31), but those months sit inside the liquidity holdout, so the researcher had already watched 2026. What was not done: no coefficient was fit on the held-out returns, and no headline was chosen because the held-out Sharpe was the largest of a grid. What was done: two later hypotheses were written by someone who had already seen that this particular two-year window paid the high-VIX state. Treat A's out-of-sample slope as the clean confirmation. Treat B's +0.24 and C's rejection as pre-registered follow-ups on a window that was no longer sealed.